

KNUST Attendance

QR-based Attendance System — Complete User Manual

Built for PESA KNUST · Kwame Nkrumah University of Science and Technology

Version	1.0
Audience	Lecturers, Teaching Assistants, Students
Live URL	https://knust-qr-scan.lovable.app

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1. Overview & Concepts

The KNUST Attendance app replaces paper sign-in sheets with a secure, real-time QR code system. Every lecturer creates their own **master admin account**, uploads their students once, and marks attendance via one of three methods:

- **Manual scan** — the lecturer/T.A. scans a student's personal QR code from their phone.
- **Projected QR self check-in** — the lecturer projects a session QR; students scan it and confirm their identity + location.
- **Master Portal link** — students retrieve their personal QR from a single shared portal link.

Each lecturer only sees the data they created; other lecturers' students are invisible.

2. Getting Started — Creating an Account

- Open the app: <https://knust-qr-scan.lovable.app>
- Click **Sign in** then choose **Continue with Google**, or register with an email + password.
- Every new account is automatically granted the **Master Admin** role for its own data.
- After signing in you land on the **Dashboard**.

3. Dashboard

The dashboard shows quick counts (students, courses, open sessions) and today's scans. Use the sidebar to open any feature.

4. Departments & Academic Years

- Add every department/programme your students belong to (name + short code).
- Departments can be **renamed** or **deleted** inline.
- Add an academic year (e.g. **2025/2026**) and click **Set current** to mark the running year.
- Academic years can also be renamed or removed.

5. Courses & Rosters

- Create a course with code, title, level, semester, department and academic year.
- Open the **Roster** to enrol students into that course. Students who scan for any session of the course are auto-registered.
- Edit or delete a course any time.

6. Students — Import, Manage, Group by Level

6a. Add one at a time

Fill the form on the Students page (name, index number, level, email, department).

6b. Bulk import from Excel

- Click **Import students** and pick your .xlsx / .csv file.
- Recognized columns (case-insensitive): **Name / Full Name**, **Index Number / Matric**, **Level / Year**, **Email / Gmail**, **Department / Programme**.
- If a department column is missing, the **Programme** column is used as the department name.
- Duplicates by index number are skipped automatically.

6c. Import emails only

Use **Import emails** to attach Gmail addresses to already-imported students. Names are matched by token-set so 'John Doe' matches 'Doe John'.

6d. Grouping

Students are grouped by level (100 / 200 / 300 / 400) via tabs. The **All** tab shows count-per-level headers.

6e. Delete all

The **Delete all** button clears every student you own after a two-step confirmation.

7. Sessions — Creating a Class Session (with GPS Geofence)

- Open **Sessions** → **New session**.
- Pick the course, optionally give it a title (e.g. *Week 4 Lecture*).
- Tap **Use my current location** to capture the classroom's GPS coordinates.
- The default geofence radius is **80 metres** — students outside this circle cannot self check-in.
- Grace minutes control what counts as being 'on time' — scans after *starts_at* + *grace* are marked **Late**.
- Sessions automatically close after 12 hours; you can also close them manually from the Scanner page.

8. Scanner — Taking Attendance

- Choose an open session on the Scanner page.
- Grant camera access and point at each student's personal QR code.
- Duplicate scans within 30 minutes are blocked (prevents accidental double-check-in).
- 'Recent scans' resets every 24 hours.
- Use **Close session** to end the session and auto-mark all non-scanners as absent.

9. Projected QR — Students Self Check-In

- On the Sessions page, click the **Projector** icon of an open session.
- The screen displays a large QR for students to scan from their seats.
- The QR points to the **public** URL, so it works even if students aren't signed in.
- Each student enters their index number and allows location access. If they are within the geofence, they are marked **PRESENT**. Otherwise they see how far away they are.

10. Student QR Portal (Single Shared Link)

- Open **Student Portal** in the sidebar.
- There is one **permanent shared link** for your class. Copy it and give it to students — the same link works for every course you teach.
- On the portal, students first pick the course, then enter their index number + registered email to retrieve their QR code (downloadable as PNG or PDF).
- Only students you own can look themselves up — other lecturers' students are invisible.

11. Reports — Daily & Semester, with Lateness

- Choose a course, then a view: **Overall (semester)** or **Daily (one session)**.
- Each session column is labelled with its date, week number (**W1, W2...** counted from the first session of the course), and optional title.
- Cells show **1** for on-time scans, **L** for late scans, and **0** for absent.
- The **Total** column counts on-time + late scans. A **Late** column shows total lateness.
- Export the current view to **Excel**, **CSV** or **PDF**. Filenames include the course and mode.

12. Anti-Cheating & Security Model

- **Rotating session QR** — each session has its own QR bound to that session only.
- **GPS geofence (80 m default)** — students must physically be inside the classroom circle.
- **High-accuracy positioning** — the browser requests GPS with high accuracy (not IP location).
- **One record per session per student** — a database unique constraint prevents duplicates.
- **30-minute lockout** on the manual scanner prevents rapid re-scans of the same code.
- **Server-side validation** — check-in runs inside a SECURITY DEFINER function; the client cannot bypass distance checks.
- **Row-Level Security** — every table is scoped by `owner_id = auth.uid()`. Lecturers cannot read each other's students, sessions, or reports.

13. Data Ownership & Privacy

- Every record you create is tagged with your **owner_id** and only visible to you.
- Students' emails are used only to let them retrieve their own QR — never sold, shared, or exposed to other lecturers.
- You can delete your entire student list at any time from the Students page.
- The backend is hosted on Supabase (encrypted at rest, TLS in transit).

14. Troubleshooting & FAQ

The camera won't open on the Scanner page

Grant camera permission in your browser settings. On iOS use Safari; on Android use Chrome.

A student says 'You appear to be Nm away'

They are outside your geofence. Ask them to move into the classroom, or increase the radius when creating the session.

Excel import ignored a column

Rename the column header to one of the recognized names in section 6b.

Portal link says 'Invalid or inactive link'

The lecturer disabled the link. Re-enable it from Student Portal.

Session shows CLOSED but I didn't close it

Sessions auto-close 12 hours after their start time.

15. Terms of Service (Summary)

- The app is provided **as-is** for academic attendance tracking. No warranty of uptime or fitness for a particular purpose.
- You are responsible for the accuracy of the student data you upload and for informing your students how attendance is tracked.
- You must not upload data you do not have the right to process, or use the system to harass or deceive students.
- You must comply with your university's data-handling policies and Ghana's Data Protection Act, 2012 (Act 843).
- We may suspend accounts that abuse the service, distribute spam, or attempt to bypass its security features.
- Full terms are available at **/terms** in the web app.

16. Privacy Policy (Summary)

- **Data collected:** lecturer email + name (from Google/email sign-up); student name, index number, level, department, email and QR UUID uploaded by the lecturer; attendance timestamps and coarse GPS coordinates during check-in.
- **How used:** to authenticate lecturers, generate personal QR codes, verify classroom presence, and produce attendance reports for the owning lecturer.
- **Retention:** data is kept until the owning lecturer deletes it. Deleting a course/session/student removes the associated records.
- **Sharing:** we do not sell data. Data is stored on Supabase infrastructure and processed only to run the service.
- **Your rights:** students may contact their lecturer to request access, correction or deletion of their record.
- Full policy is available at **/privacy** in the web app.